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Direct mapping of tyrosine sulfation states in native peptides by nanopore

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Sulfation is considered the most prevalent post-translational modification (PTM) on tyrosine; however, its importance is frequently undervalued due to difficulties in direct and unambiguous determination from phosphorylation. Here we present a sequence-independent strategy to directly map and quantify the tyrosine sulfation states in universal native peptides using an engineered protein nanopore. Molecular dynamics simulations and nanopore mutations reveal specific interactions between tyrosine sulfation and the engineered nanopore, dominating identification across diverse peptide sequences. We show a nanopore framework to discover tyrosine sulfation in unknown peptide fragments digested from a native protein and determine the sequence of the sulfated fragment based on current blockade enhancement induced by sulfation. Moreover, our method allows direct observation of peptide sulfation in ultra-low abundance, down to 1%, and distinguishes it from isobaric phosphorylation. This sequence-independent strategy suggests the potential of nanopore to explore specific PTMs in real-life samples and at the omics level.

Tyrosine sulfation is the transfer of a sulfate group to the hydroxyl group of a tyrosine residue, forming a tyrosine sulfate ester[1,2]. It has a key role in protein–protein interactions and is implicated in various physiological and pathological processes, including hormonal regulation, hemostasis, inflammation and viral entry into host cells[3–8]. Despite being estimated as the most abundant post-translational modification (PTM) of tyrosine, with about 1% of the tyrosine residues being sulfated[9], only approximately 160 tyrosine-sulfated proteins have been experimentally determined over nearly 70 years of investigation. This contrasts sharply with the more than 30,000 tyrosine-phosphorylated and tyrosine-glycosylated proteins[10]. Therefore, tyrosine sulfation remains one of the most undercharacterized PTMs, which needs a direct and efficient approach for accurate measurement.

Measurement of tyrosine sulfation is challenging and relies on limited approaches, mainly via isotope labeling, antibodies, fluorescence labeling and mass spectrometry (MS)[11]. Isotope labeling and fluorescence labeling require complex and time-consuming procedures and sometimes can result in ambiguous determination of sulfation presence and position. Antibodies often lack sensitivity and generality. Despite MS being a powerful tool for exploration of PTMs, the determination of tyrosine sulfation and its site remains demanding. First, tyrosine sulfation is sensitive to low pH and thermal degradation, leading to the neutral loss of sulfur trioxide (SO_3) during collision-induced dissociation[10,12]. Second, phosphorylation and sulfation have similar chromatographic properties and differ in mass by only 9.5 mDa[12], presenting a substantial challenge for MS in directly identifying them, particularly in large peptides/proteins, even when using high-resolution MS[13]. Chemical labeling to normal tyrosine was developed to indirectly determine tyrosine sulfation by using the instability of the sulfate group in MS analysis[14]. However, these approaches are indirect and expensive, limiting the accurate and efficient mapping of tyrosine sulfation on native proteins at a large scale or even at the omics level. More importantly, detecting PTMs on proteins often suffers from their low concentration and abundance[15]. For example, the total concentration of tyrosine-sulfated proteins in human plasma is only around the nanogram per milliliter level[16], let alone the concentration of each protein.

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Proteins with low abundance are often lost during bulk measurement, further hindering PTM determination. Therefore, a high-throughput single-molecule approach is urgently needed for the direct and unambiguous determination of tyrosine sulfation states.

Nanopore was developed for single-molecule DNA sequencing three decades ago and now has been attracting increased attention for protein sensing and sequencing, including protein PTM detection[17–27]. By characterizing individual molecules, nanopore inherently reveals the heterogeneity between and within molecules and avoids interference from bulk properties, even at ultra-low abundance. Recent studies successfully applied nanopore to identify various PTMs, such as phosphorylation, acetylation and their modification sites[19,21,24]. However, these cases highly depend on peptide sequences and/or the regulated transport of the peptides supported by molecular motors or special structures. Here we propose a sequence-independent strategy for directly mapping tyrosine sulfation states in universal native peptides without interfering with tyrosine phosphorylation. By enhancing the driving forces for peptide entry and introducing an energy barrier for peptide stabilization inside the nanopore through the T232K mutation, the specific interactions between tyrosine sulfation and aerolysin sensing region were improved and used to determine the sulfation states in a decapeptide from C-C chemokine receptor type 2 (CCR2). All-atom molecular dynamics (MD) simulations demonstrated that sulfation significantly enhanced the non-covalent interactions through van der Waals interactions, electrostatic interactions and hydrogen bonds. These findings were further validated by the mutation in the sensing region. Accordingly, our strategy was further used to identify the sulfation states in other native peptides with entirely distinct sequences, including the fragments of P-selectin glycoprotein ligand-1 (PSGL-1), CCR5 and histone 3 (H3). Moreover, the results suggested a direct and positive correlation between the sulfation-induced blockade enhancement and the mass of peptides. Based on this, a framework was established to identify and quantify tyrosine sulfation in a peptide mixture, enabling further exploration of sulfation in a native protein. Furthermore, the determination of the sulfation on peptides was achieved at ultra-low abundance, even down to 1%. By using the specific interactions of the sulfate group, sulfation could also be clearly distinguished from isobaric phosphorylation at the same site in universal native peptides.

Results

Mapping the sulfation states with specific interactions

The nanopore signal, the ionic current blocked by the individual molecule, is affected by various features of the target molecule, including its mass, conformation and charge. These features indeed complicate and hinder the determination of particular PTMs in distinct or unknown sequences in current nanopore studies[24,26]. Non-covalent interactions between a target molecule and a nanopore have been demonstrated to combine with volume exclusion to determine the current blockade[28], enhancing the ability of the nanopore for single-unit discrimination. Therefore, the specific non-covalent interactions between the sulfate group and nanopore were designed and used to achieve a generalized response to tyrosine sulfation on universal peptide sequences. Owing to the negative charge and the hydrophilicity of the sulfate group, the electrostatic interactions and the hydrogen bonds between tyrosine sulfation and nanopore are key factors for specifically amplifying the change in current blockade. Notably, the large aromatic side chain of tyrosine shortens the distance between the sulfate group and the side chains of nanopore lumen, simultaneously enhancing the electrostatic interactions and increasing the probability of hydrogen bonds formation. Accordingly, aerolysin nanopore with a sensing region containing high concentrations of charged amino acids (for example, Arg220 and Asp222) and hydrophilic amino acids (for example, Arg220, Thr274 and Ser276) (Fig. 1a and Supplementary Fig. 1) was chosen as the candidate for sulfation detection. Given the roles of tyrosine sulfation in the activation and downstream signaling of CCR2 in immune and inflammatory responses[29], a decapeptide derived from CCR2 (21-TTFFDYDYGA-30, $CCR2_{21–30}$) that contains the tyrosine sulfation motif was taken as an example to evaluate the ability of the nanopore to detect tyrosine sulfation (Fig. 1a).

As shown in Supplementary Fig. 2a, in wild-type aerolysin, the unmodified $CCR2_{21–30}$ induced only short-lived current events with a wide distribution in current blockade (I/I_0). Although sulfated peptides could not be well discriminated, a clear prolongation in duration and a narrower distribution in current blockade for the sulfated peptides were observed (Fig. 1c and Supplementary Fig. 2b–d). These observations demonstrated the expected enhancement in non-covalent interactions between tyrosine sulfation and the aerolysin sensing region. Given the diverse charges of native peptides and the significance of long duration for improving the ability of a nanopore to discriminate subtle differences[30,31], the positively charged lysine with a long side chain was introduced into the non-sensitive region of aerolysin. This mutation aimed to simultaneously enhance the electroosmotic flow (EOF) within nanopore and create an energy barrier for peptide translocation, thereby prolonging the duration of the peptide in the sensing region. Thus, a series of lysine mutations along the non-sensitive region of aerolysin was performed (Fig. 1b and Supplementary Figs. 3–5). In all mutant aerolysin nanopores, the introduction of the lysine resulted in higher event frequencies and longer duration (Fig. 1c and Supplementary Figs. 6–11), demonstrating the strengthened EOF inside nanopore and the successful construction of the designed energy barrier. However, despite the nearly same diameters for the designed mutation sites based on a 2.3 Å resolution cryo-EM structure of aerolysin (Fig. 1b and Supplementary Fig. 5), the lysine mutation at different positions varied in their ability to detect sulfation. As illustrated in Fig. 1b, as the distance of the mutation from the sensing region (near Arg220) increased, the duration of the peptides was significantly prolonged. Accordingly, the G270K and S228K mutation near the sensing region exhibited extremely low resolution for sulfation detection (Fig. 1c and Supplementary Figs. 6 and 7), with the duration of sulfated peptides similar to or even shorter than that of the unmodified peptide. These results suggested that the lysine mutation would significantly affect the response of the neighboring sensing region to tyrosine sulfation. As shown in Fig. 1c and Supplementary Fig. 8, in T230K mutation with a relatively longer distance, the prolonged duration induced by tyrosine sulfation was reobserved. In addition, the T230K mutant aerolysin could identify and quantify sulfation in $CCR2_{21–30}$, although it could not yet discriminate the sulfation site. Moreover, with the distance further increasing, T232K or S264K mutant aerolysin nanopores could fully determine the tyrosine sulfation and its sites (Fig. 1c and Supplementary Fig. 9). This indicates that the introduction of lysine should avoid disturbing the readout for tyrosine sulfation in the sensing region. The decrease in the duration of CCR2 peptides in S264K mutant aerolysin could be ascribed to the shortened distance from another charged region in the aerolysin bottom (Supplementary Figs. 1 and 11). Thus, the T232K mutant aerolysin, which introduces lysine into the middle of the non-sensitive region, exhibits the most optimized ability for tyrosine sulfation detection.

To further assess the capabilities of the T232K mutant aerolysin, sequential addition experiments were performed. As illustrated in Fig. 1d, the addition of sulfated $CCR2_{21–30}$ at Tyr_{26} ($CCR2_{21–30}$-Y26s) resulted in characteristic current events that were nearly 12 times longer in duration and approximately 0.64-pA deeper in current blockade compared to unmodified $CCR2_{21–30}$ at +50 mV (Supplementary Fig. 12). This led to the well-separated populations on the scatter plots (Fig. 1c,d). More interestingly, despite owning equal mass and charge (Fig. 1a), the sulfation at the other tyrosine residue on $CCR2_{21–30}$ ($CCR2_{21–30}$-Y28s) could be clearly distinguished from $CCR2_{21–30}$-Y26s, showing a blockade that was nearly 1.37-pA deeper (Supplementary Fig. 12b). The deepest current blockade was observed with the addition of the sulfated $CCR2_{21–30}$ at both tyrosine residues ($CCR2_{21–30}$-Y26/28s) (Fig. 1c,d). These results demonstrated that the designed T232K mutant aerolysin could indeed directly determine the tyrosine sulfation states

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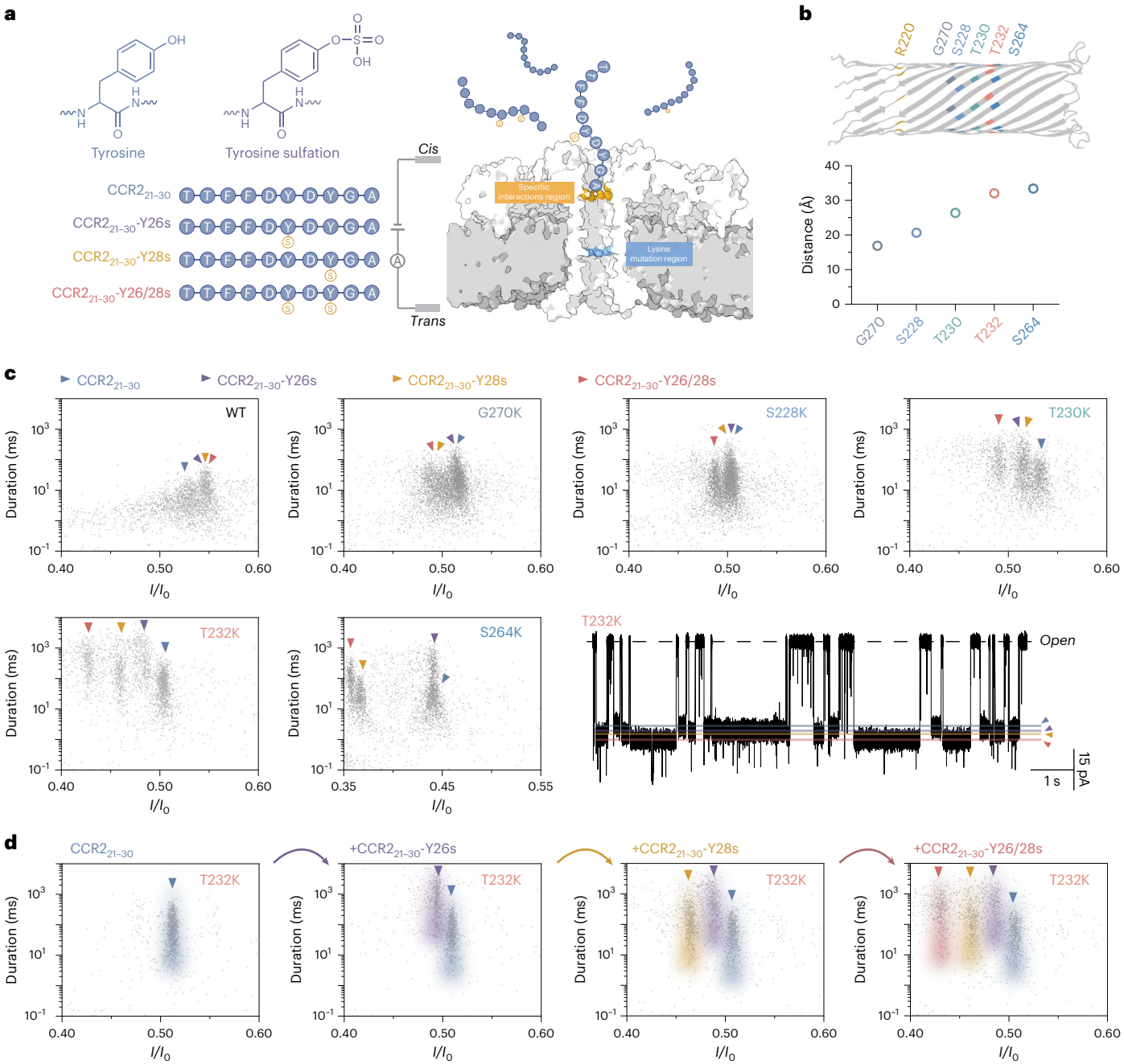


Fig. 1 | Single-channel recording for mapping the sulfation states of tyrosine with mutant aerolysin nanopore. a, Illustration of the peptide sulfation and the nanopore detection system. The chemical structure of tyrosine, sulfated tyrosine and the sequence of CCR2$_{21-30}$ peptide are shown here. The yellow circles indicate the positions of tyrosine residues (Tyr26 and Tyr28) that are susceptible to PTM by sulfation. Aerolysin nanopore was inserted in a lipid bilayer membrane, which separated the detection system into *cis* chamber and *trans* chamber. **b**, The relative distance from the amino acid sites on the non-sensitive region to the Arg220 site of the sensing region of aerolysin. The calculation is based on the vertical distance of the center points of the planes formed by seven amino acid residues, computed using VMD software[42]. **c**, Scatter plots of duration time versus I/I_0 at +60 mV based on WT and a series of mutant aerolysin nanopores with a mixture of four peptides, CCR2$_{21-30}$ and its sulfated peptidoforms. The peaks of each peptide are marked with triangles corresponding to different colors. The current trace with the T232K mutant aerolysin is shown here. **d**, Scatter plots of duration time versus I/I_0 at +60 mV with the sequential addition of CCR2$_{21-30}$, CCR2$_{21-30}$-Y26s, CCR2$_{21-30}$-Y28s and CCR2$_{21-30}$-Y26/28s. Data were acquired using 1.0 M KCl, 10.0 mM Tris buffer at pH 8.0 and 24 ± 2 °C, unless otherwise specified. The measurements and mutants were performed as described in the Methods.

on CCR2$_{21-30}$. Although the extra negative charge of the sulfate group could facilitate peptide translocation, the sulfation significantly prolonged the duration of the peptide inside nanopore. Thus, these results further validate the enhanced interactions between T232K mutant aerolysin and sulfated peptides as designed, indicating that the T232K mutant aerolysin could potentially map tyrosine sulfation states in universal peptides.

Interaction-determined mechanism revealed by MD simulations

To validate and further elucidate the sulfation-enhanced interactions within the T232K mutant aerolysin, all-atom MD simulations were constructed using NAMD[32] to gain deeper insights into the molecular mechanism underlying the determination of sulfation states. As shown in Fig. 2a, a full-length aerolysin nanopore with T232K mutation was

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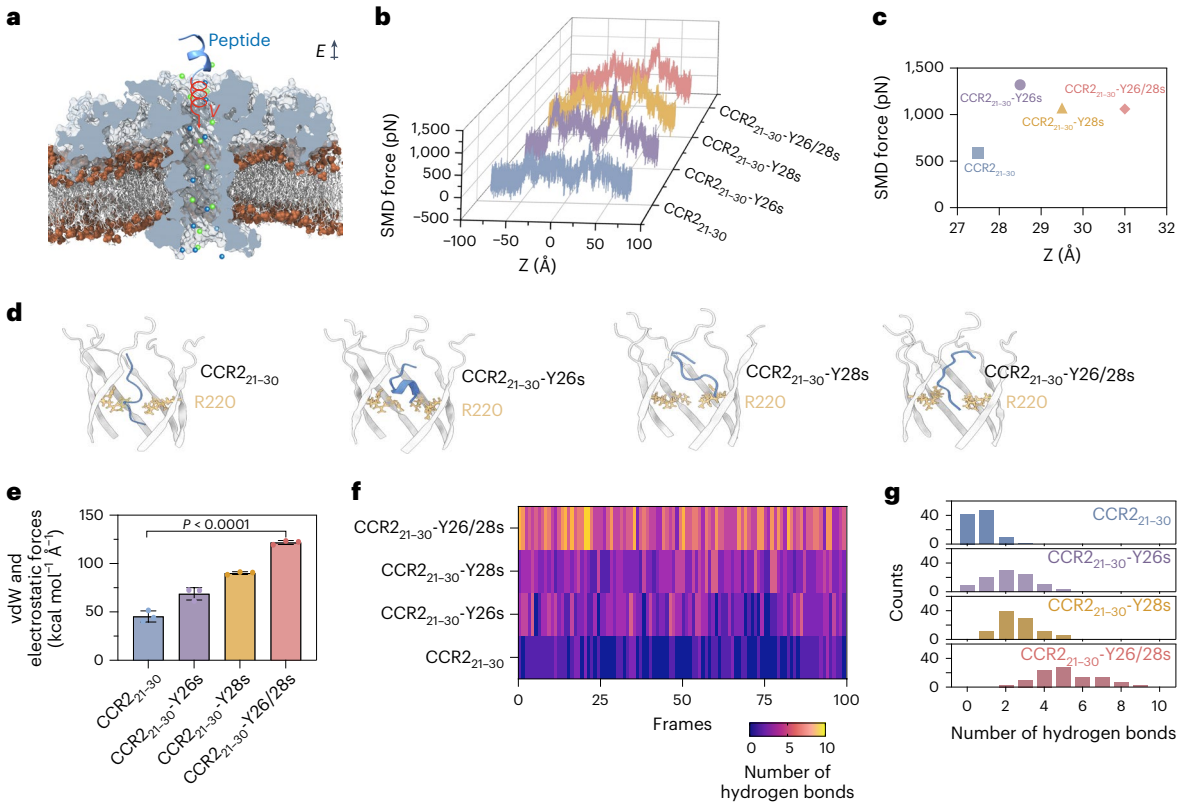


Fig. 2 | The molecular mechanism of T232K mutant aerolysin mapping the tyrosine sulfation states by MD simulations. a, All-atom model of the full-length aerolysin nanopore system with T232K mutation. The mutant aerolysin nanopore is embedded in a lipid bilayer and submerged in 1.0 M KCl (only the ions inside aerolysin are shown). The model of CCR2$_{21-30}$ was truncated from the AlphaFold prediction model (Supplementary Fig. 13). The peptide was placed at the *cis* side of the T232K aerolysin and then pulled through the nanopore at a constant velocity. **b**, The SMD force exerted on the peptidoforms at different positions for the transport. **c**, The maximum SMD force and the corresponding positions of peptidoforms. **d**, Illustration of each peptidoform constrained at the maximum SMD force position as shown in **c**. **e**, van der Waals (vdW) and electrostatic forces between peptidoforms and nanopore based on 2-ns MD trajectory within the final 20-ns simulation; the error bar here is from three independent simulations. Data are presented as mean ± s.d. for $n = 3$ replicates. Statistical analysis was conducted using ordinary one-way ANOVA with Tukey's multiple comparison test, and P values are shown. $P < 0.05$ was considered statistically significant. **f**,**g**, The number of hydrogen bonds between peptidoforms and nanopore at every 20 ps during the same time period of the calculation of van der Waals and electrostatic interactions (**f**) and the corresponding histograms (**g**). The hydrogen bonds were determined by geometry criterion (Methods).

constructed and inserted into a lipid membrane, surrounded by a water box containing 1.0 M KCl to replicate the experimental conditions. The model of CCR2$_{21-30}$ was cut from the full-length CCR2 structure predicted by AlphaFold[33]. The sulfation models of CCR2$_{21-30}$-Y26s, CCR2$_{21-30}$-Y28s and CCR2$_{21-30}$-Y26/28s were generated by patching a sulfate group to the specific tyrosine (Methods and Supplementary Fig. 13). It was demonstrated that the current blockade induced by a peptide inside aerolysin is greatly correlated to its location within the nanopore lumen[21,34]. The constant current blockade value was maintained throughout the entire duration for all fragments (Fig. 1c), which suggests that the fragments were immobilized in specific regions during most of their translocation. Therefore, to determine the region of the peptides staying, steered molecular dynamics (SMD) simulations were used to simulate the translocation features, by pulling CCR2 fragments through the pore at a constant velocity of 0.005 Å ps^{-1}, based on previous studies[28,34] (Fig. 2a and Methods). The same CHARMM36 force field was chosen to ensure comparability of results[28,34,35]. According to the experimental results, the entry of CCR2$_{21-30}$ peptides is primarily driven by the enhanced EOF within the T232K mutant aerolysin, along with the electrophoretic force acting on the negative charge groups (sulfate group and aspartic acid residue) in the peptide sequence. Thus, to accurately model the forces exerted on the peptides, the backbone of the peptide, along with the side chain of the negatively charged amino acids and the sulfate group on tyrosine, were attached to a dummy atom to facilitate translocation (model details provided in the Methods). The SMD force here represents the force along with the traverse orientation to overcome the energy barrier from the steric hindrance and the non-covalent interactions between peptides and aerolysin. As shown in Fig. 2b, the SMD force increases rapidly as soon as the peptides entered aerolysin, especially at 27.5 Å for CCR2$_{21-30}$, 28.5 Å for CCR2$_{21-30}$-Y26s, 29.5 Å for CCR2$_{21-30}$-Y28s and 31 Å for CCR2$_{21-30}$-Y26/28s (Fig. 2c). Because the position where the molecules stay has been demonstrated to greatly affect the current blockade, these results suggest that the differences in the position of peptides could be another crucial factor for the discrimination of the four peptides. In addition, sulfation of the peptide results in nearly two-fold enhancement in the highest energy barrier, with CCR2$_{21-30}$, CCR2$_{21-30}$-Y26s, CCR2$_{21-30}$-Y28s and CCR2$_{21-30}$-Y26/28s corresponding to the SMD force of 588 pN, 1,320 pN, 1,072 pN and 1,061 pN, respectively (Fig. 2c). Moreover, to further validate these observations, the attachment to the dummy atom was changed to the backbone of the peptides, considering only the EOF as the driving force due to the potential effect of the counter ions binding to the negative charges of the peptides[36,37]. The results showed that, even with the change in forced atoms, significantly stronger SMD forces were still obtained in sulfated peptides (Supplementary Fig. 14), demonstrating a higher energy barrier induced by tyrosine sulfation. Additionally, the largest SMD force for each peptide was also observed in the sensing region, although there was a slight shift in the precise

position. This shift could be ascribed to the relaxation of the charged side chain and the sulfation group, affecting the interactions with the nanopore. The significantly enhanced energy barriers induced by sulfation observed in MD simulations corresponded with their prolonged duration seen in the experiments as presented in Fig. 1. This confirms the reliability of the simulations and the strong response of the mutant nanopore to tyrosine sulfation.

The non-covalent interactions between each CCR2 peptide and the nanopore were studied to further reveal the mechanism of the response to peptide sulfation. Here, the peptide was stabilized at the position where the SMD force was highest as illustrated in Fig. 2d (Methods). The calculation of non-covalent interactions, including van der Waals interactions and electrostatic interactions in the NAMD Energy plugin (Methods), showed that the sulfation on tyrosine could directly enhance these interactions between the peptides and the T232K mutant aerolysin. The enhancement is particularly significant for $CCR2_{21-30}$-Y26/28s, with interactions nearly 2.7 times stronger than those of $CCR2_{21-30}$ (Fig. 2e and Supplementary Fig. 15). More interestingly, $CCR2_{21-30}$-Y28s exhibits about 1.3-fold the van der Waals and electrostatic forces than the same charged $CCR2_{21-30}$-Y26s. This supports the notion that the negative charge of the sulfate group in $CCR2_{21-30}$-Y28s was greatly paired, leading to a low effective charge as calculated from the voltage-dependent experiments (Supplementary Table 2 and Supplementary Fig. 12a). As shown in Fig. 2f and Supplementary Fig. 16, the numbers of hydrogen bonds between CCR2 peptides and the mutant nanopore were calculated by using geometry criterion (calculation details provided in Methods). The observations suggested more obvious changes in hydrogen bonds induced by sulfation states, with nearly fivefold and tenfold enhancements for single and double sulfation, respectively (Fig. 2f,g). Despite the presence of negatively charged and hydrophilic amino acid residues in $CCR2_{21-30}$, sulfation on the peptides still contributed to most of the non-covalent interactions with nanopore, including van der Waals interactions, electrostatic interactions or hydrogen bonds. These results were confirmed by MD simulations using the latest CHARMM36m force field (Supplementary Fig. 17). Therefore, the non-covalent interactions indeed have critical roles in determining the sulfation states of peptides. To further validate the non-covalent interaction-dependent mechanism, with the T232K mutation maintained, the Arg220 was mutated because it was considered to greatly contribute to both electrostatic interactions and hydrogen bonds with tyrosine sulfation. As illustrated in Supplementary Fig. 18, the mutation on Arg220 by R220S/T232K mutant aerolysin severely impaired the ability for sulfation detection. Moreover, even by replacing Arg220 with glutamine, which has a large side chain to minimize the diameter change, the R220Q/T232K mutant aerolysin also could not distinguish tyrosine sulfation (Supplementary Fig. 19). By contrast, the mutation in Asp222 (D222S/T232K) only slightly decreased the duration of each peptide, whereas its ability to identify all four sulfation states of CCR2 peptides was almost not affected (Supplementary Fig. 20). These results highlighted the importance of non-covalent interactions between tyrosine sulfation and aerolysin sensing region, especially for Arg220, in sulfation detection, consistent with observations from MD simulations. Additionally, although the non-covalent interactions for $CCR2_{21-30}$-Y26s were lower than those for $CCR2_{21-30}$-Y28s, the higher total energy barrier for $CCR2_{21-30}$-Y26s could be ascribed to its distinct conformation that adopts a helix in nanopore confinement as illustrated in Fig. 2d. Therefore, these results suggested the non-covalent-interaction-specific mechanism for sulfation state measurement and demonstrated the potential of the designed nanopore to map the sulfation states independently of the peptide sequence.

Identifying the sulfation states in universal native peptides

Based on the unique non-covalent interactions demonstrated above, we further validated the specific response of our strategy to sulfation in other native peptides, even with entirely different sequences. Here, a short fragment (42-QATEYEYLDY-51, $PSGL_{42-51}$) near the amino terminus of PSGL-1 was chosen, where the sulfation has been known to control the binding activity with P-selectin[3,4]. Although $PSGL_{42-51}$ also contains a sulfation motif similar to $CCR2_{21-30}$, its amino acid sequence in other parts are quite different. As shown in Fig. 3a, with the addition of the PSGL fragment containing the sulfation in Tyr46 ($PSGL_{42-51}$–Y46s), a new population gradually appeared on the scatter plots, which could be well separated from the unsulfated $PSGL_{42-51}$. Moreover, the sulfation on the neighboring tyrosine ($PSGL_{42-51}$–Y48s) could also be determined, which demonstrated the specific response of the sequence-independent strategy to sulfation. The current blockades corresponding to PSGL peptides in the mixture were verified in their pure samples (Supplementary Fig. 21). To further evaluate the effects of the neighboring amino acids on sulfation discrimination, we tested two other fragments from CCR5 (ref. 38) and H3 (ref. 39), which own entirely distinct sulfation motifs with the sequence of 7-SPIYDINYYT-16 ($CCR5_{7-16}$) and 98-AYLVGLFEDT-107 ($H3_{98-107}$), respectively. The results showed that, even with neutral amino acids nearby, the tyrosine sulfation, as well as its site, could be determined in the CCR5 fragment (Fig. 3b and Supplementary Fig. 22). Moreover, with the sulfation identified on $H3_{98-107}$ (Fig. 3c), the impurity presence of unsulfated $H3_{98-107}$ in pure $H3_{98-107}$–Y99s sample could be clearly observed (Supplementary Fig. 23), which could be ascribed to the loss of sulfation group during storage. All these observations demonstrated that the neighboring amino acids would hardly affect the accurate identification of tyrosine sulfation in diverse native peptides. These results further support the specificity of the interactions between tyrosine sulfation and the nanopore, which are not disturbed or hindered by the peptide sequence. Notably, the applied voltage would slightly affect the resolution of the nanopore and the efficiency for sulfation detection due to the effect on peptide duration and frequency (Supplementary Figs. 24 and 25). Therefore, to achieve optimal identification of tyrosine sulfation in universal native peptides or unknown peptide mixtures, it is recommended to test a range of applied voltages during the nanopore experiments.

Although computational and experimental studies have shown a sequence-independent response to tyrosine sulfation, a hidden correlation between the peptides and sulfation-induced current enhancement was further explored. According to the previous studies on the origin of nanopore events, the current blockade could be divided into two parts: the volume exclusion, which affects the space for ion transport, and the interactions influencing the mobility of the ions. Thus, the enhancement in current blockade induced by sulfation could be simplified as:

$$\Delta I_{\text{sulfation}} = f(V_s)\cdot f(I_s) - f(V)\cdot f(I) \approx f(V)\left(f(I_s) - f(I)\right)$$

where V_s and V represent the volume exclusion of the sulfated and unmodified peptide, respectively, and I_s and I represent nanopore interactions formed with the sulfated and unmodified peptide, respectively (Methods). Because the sulfation has been demonstrated to contribute most of the interactions with the nanopore, $f(I_s) - f(I)$ could be considered as a constant value. Volume exclusion is mainly controlled by the mass and the conformation of the analyte inside nanopore[40]. By roughly ignoring the potential effects of conformation as the nanoconfinement of aerolysin (~1 nm) restricting most protein/peptide conformation, the enhanced current blockade induced by sulfation should be directly correlated to the mass of the peptides. As expected, a positive linear correlation was observed between the mass of the peptides and the average blockade enhancement induced by tyrosine sulfation in all tested sulfated peptides (Fig. 3d). The slight shift in the current blockade enhancement for sulfation at different sites on a particular peptide could be attributed to the distinct positions where the peptide resides or to the potential different structures induced by the sulfation. Moreover, the di-sulfation would result in about two-fold enhancement compared to the mean value of the single sulfation in the same sequence (Supplementary Fig. 26), verifying the

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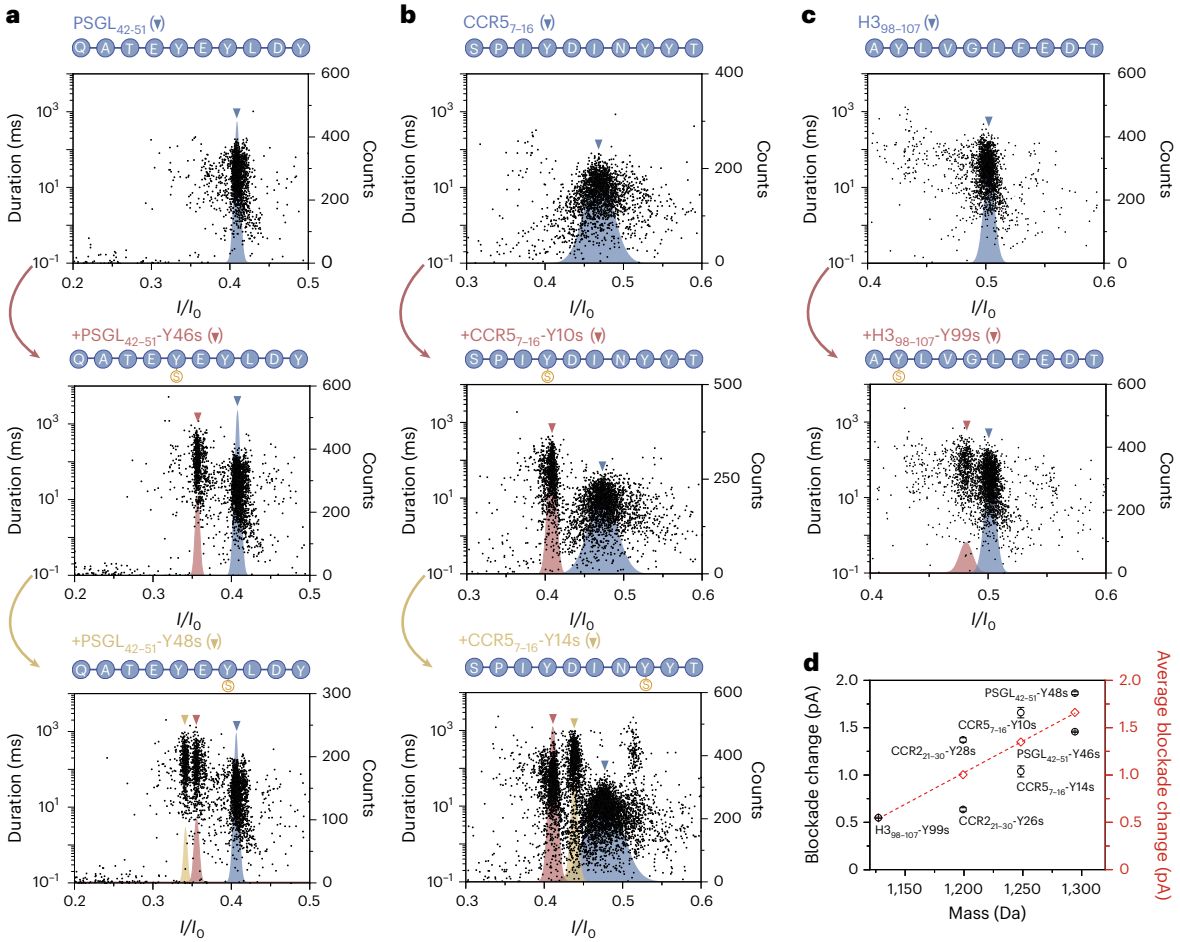


Fig. 3 | Determining the tyrosine sulfation without the influences of peptide sequences. a, Scatter plots and corresponding histograms of the sequential addition of $PSGL_{42-51}$ (top), $PSGL_{42-51}$–Y46s (middle) and $PSGL_{42-51}$–Y48s (bottom). **b**, Scatter plots and corresponding histograms of the sequential addition of $CCR5_{7-16}$ (top), $CCR5_{7-16}$–Y10s (middle) and $CCR5_{7-16}$–Y14s (bottom). **c**, Discrimination of $H3_{98-107}$ and $H3_{98-107}$–Y99s. **d**, The correlation between the peptide mass (Supplementary Table 1) and the blockade enhancement induced by sulfation with the corresponding linear fitting shown in red. The fitting was based on the mass of unsulfated peptides and the average blockade enhancement of the sulfation at different sites (Methods). Data are presented as mean ± s.d. and from at least three independent experiments. All data were obtained at the applied voltage of +50 mV.

correlation observed above. This discovery further supported the interaction-dependent mechanism for determining tyrosine sulfation as designed and demonstrated and could be used to predict the mass of unknown sulfated peptides.

Exploring tyrosine sulfation in a native sulfated protein

Based on the sequence-independent strategy for tyrosine sulfation determination, we investigated the potential of nanopore to accurately and efficiently determine the protein/peptide sulfation in unknown samples. Notably, compared to the MS experiments, especially for MS/MS analysis, the nanopore experiments are conducted at room temperature and pH 8.0, conditions under which tyrosine sulfation is expected to be stable. To validate this, a long-period measurement was conducted. Here, taking the detection of pure sample $CCR2_{21-30}$-Y28s as an example, a tiny impurity of $CCR2_{21-30}$ was observed from the nanopore detection. This impurity may have resulted from the degradation of sulfated tyrosine during storage or from impurities remaining after purification by high-performance liquid chromatography with mass spectrometry (HPLC-MS) (Supplementary Figs. 27 and 28). The event ratio of $CCR2_{21-30}$-Y28s and $CCR2_{21-30}$ remained consistent with a deviation rate lower than 0.6% during over 12-h measurements, demonstrating the stability of tyrosine sulfation under nanopore conditions (Supplementary Fig. 29). In addition, the effects of buffer pH (2.0-10.0) on the discrimination ability and peptide sulfation stability were tested. As illustrated in Supplementary Figs. 30 and 31, although the low pH, particularly lower than 4.0, would partially hinder the sulfation detection due to the shortened duration, it hardly affected the sulfation stability. Thus, peptide sulfation remained stable and was not altered or lost during nanopore measurements. Accordingly, a nanopore framework was established to explore the protein/peptide sulfation by comparing the current events before and after peptide desulfation. Exploiting the instability of tyrosine sulfation, a desulfation procedure involving acidification and heating was designed to remove potential tyrosine sulfation on peptides (Fig. 4a and Methods). To specifically and efficiently remove sulfation, the desulfation procedure was optimized by evaluating changes in sulfated $CCR5_{7-16}$ peptides under various conditions (Supplementary Figs. 32–35). Moreover, such desulfation procedure would not influence unmodified peptides or other PTMs, confirmed by the consistent events of unmodified $CCR5_{7-16}$ and the phosphorylated peptide ($CCR5_{7-16}$–Y10p) before and after desulfation and further validated by MS analysis (Supplementary Fig. 36). Therefore, by comparing the current blockade before and after the desulfation procedure (Fig. 4a), we could determine whether and how many sulfation sites exist in the sample. Here, a peptide mixture containing sulfated $PSGL_{42-51}$–Y46s, $PSGL_{42-51}$–Y48s and $CCR5_{7-16}$–Y14s with a few unsulfated $PSGL_{42-51}$ and $CCR5_{7-16}$ was constructed to model an 'unknown' protein digested by peptidases. As shown in Fig. 4b, five distinct populations could be

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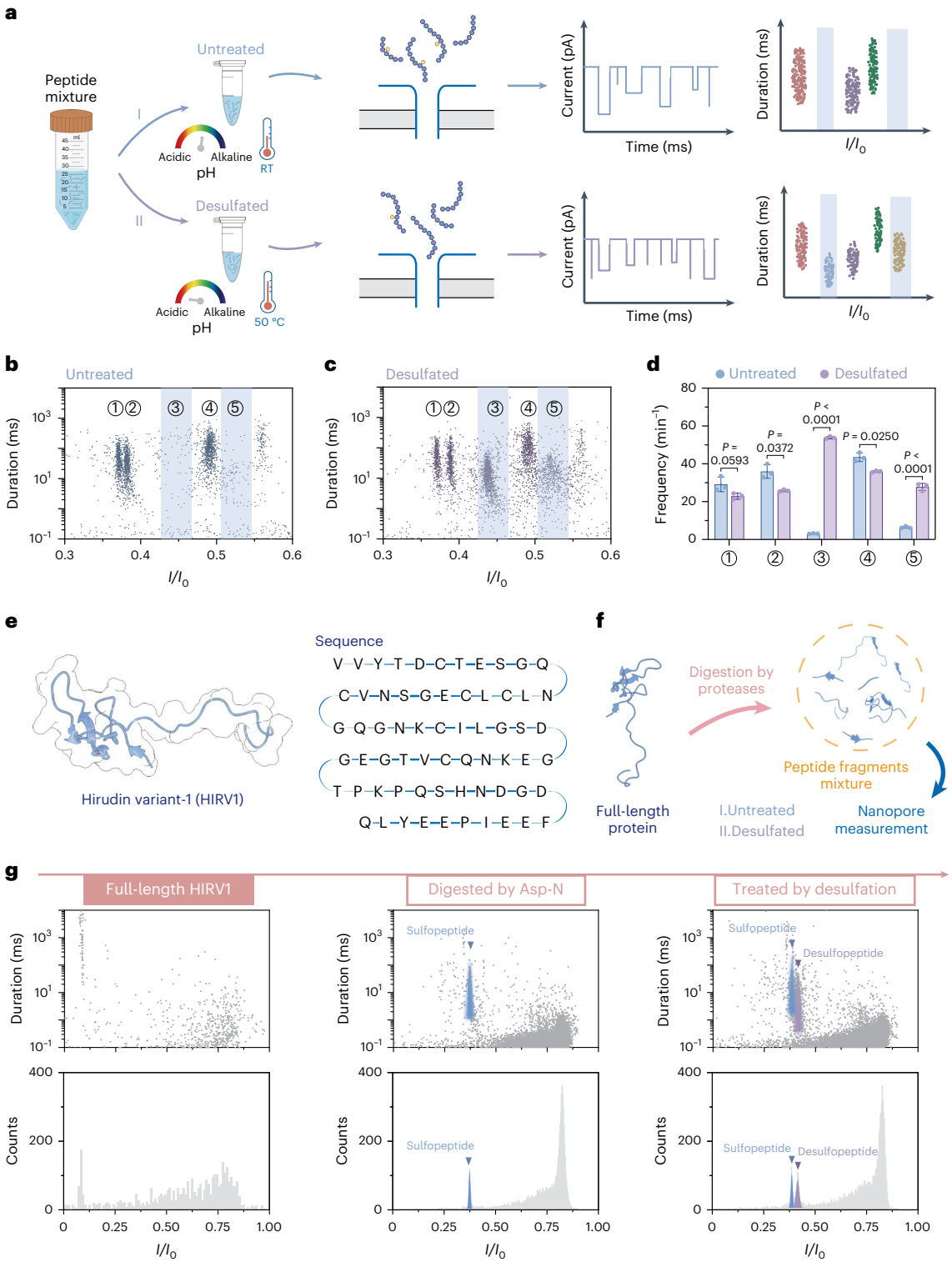


Fig. 4 | Discovering tyrosine sulfation in peptide mixtures and native proteins. a, Framework for discovering tyrosine sulfation. Procedure I: direct detection of the peptide mixture by nanopore. Procedure II: the peptide mixture is heated under acidic conditions to desulfate, after which nanopore detection is performed (Methods). Scatter plots reflect differences before and after desulfation. **b**, Scatter plot of a mixture containing $PSGL_{42–51}$–Y48s (1), $PSGL_{42–51}$–Y46s (2) and $CCR5_{7–16}$–Y14s (4), along with unmodified peptides $PSGL_{42–51}$ (3) and $CCR5_{7–16}$ (5) due to degradation or impurities. **c**, Scatter plot of the mixture in **b** after the desulfation procedure at the applied voltage of +70 mV. **d**, Changes in the event frequency of peptides before and after desulfation, presented as mean ± s.d. for $n = 3$ replicates. Statistical analysis used unpaired two-tailed t-test or Welch's t-test, with P values shown. **e**, Structure and sequence of HIRV1 protein. **f**, Illustration of discovering tyrosine sulfation in HIRV1 protein. **g**, Scatter plots and histograms of I/I_0 for full-length HIRV1 protein (left), HIRV1 peptide fragments after digestion (middle) and the digestion solution after desulfation (right) at the applied voltage of +80 mV. RT, room temperature.

observed in the untreated sample, corresponding to the five kinds of peptides. After desulfation, the event frequency of sulfated peptides ($PSGL_{42-51}$–Y48s, $PSGL_{42-51}$–Y46s and $CCR5_{7-16}$–Y14s) decreased with that of the corresponding unmodified peptides ($PSGL_{42-51}$ and $CCR5_{7-16}$) significantly increasing (Fig. 4d), suggesting that the sulfated peptides were converted to unmodified peptides. In addition, based on the standard curve of the concentration of $PSGL_{42-51}$ (Supplementary Fig. 37) and the relative capture efficiency ratio of different peptides (Supplementary Fig. 38), their concentrations were further calculated (Methods). The results showed that approximately 20% $PSGL_{42-51}$–Y48s, 27% $PSGL_{42-51}$–Y46s and 22% $CCR5_{7-16}$–Y14s were desulfated, leading to the production of approximately 1.70 μM $PSGL_{42-51}$ and approximately 0.66 μM $CCR5_{7-16}$. These results obtained from nanopore measurements were similar to the MS analysis (Supplementary Fig. 39), suggesting that the nanopore-based framework is feasible for identifying and quantifying sulfated peptides.

At present, nanopore studies on PTM discrimination require knowledge of the specific sequence being analyzed, which limits their practical applications. To directly validate the ability of the framework to explore native sulfated proteins and promote practical applications, a native sulfated protein, hirudin variant-1 (HIRV1), was detected. As illustrated in Fig. 4e, owing to the large volume of HIRV1, endoproteinase Asp-N was used to cleave it into shorter peptide fragments to facilicate their entry into nanopore. As shown in Fig. 4g, the full HIRV1 protein induced only a few long-duration events with nearly full blockades and some random bumping events. After digestion by Asp-N, more distributions were observed, corresponding to the short peptide mixture derived from HIRV1. Moreover, after the desulfation, a new distribution was observed neighboring one distribution of the untreated sample, giving a relatively smaller blockade and shorter duration (Fig. 4g). Because the desulfation procedure would not affect the peptide sequence and other PTMs, the new distribution was attributed to the desulfation of a fragment digested from HIRV1 protein, demonstrating that the HIRV1 protein contains a tyrosine sulfation site. More interestingly, based on the change in the current blockade of approximately 1.03 ± 0.03 pA, the mass of the sulfated peptide fragment was estimated to be approximately 1,202 Da. This estimation is based on the linear correlation between mass and the sulfation-induced blockade enhancement discovered earlier (Fig. 3d; calculation details provided in Methods). This estimated mass value closely matched the sequence of EEIPEEYLQ in HIRV1, which was further confirmed by MS analysis, giving the mass of 1,226.5 Da and 1,146.5 Da before and after desulfaton, respectively (Supplementary Fig. 40). Therefore, apart from determining the tyrosine sulfation on native proteins, the developed nanopore framework could further roughly estimate the mass of the sulfated fragments, which would contribute to predicting the sulfated sequence and potential site in proteins.

Considering the low abundance of tyrosine sulfation in native protein samples, we further tested the capability of the nanopore strategy to identify low-abundant sulfation. The standard curve of the concentration of the sulfated peptide $PSGL_{42-51}$–Y48s demonstrated that the limit of detection (LOD) of the nanopore for tyrosine sulfation could reach approximately 0.01 μM (~14 ng ml^{-1}). Because the occupancy of the sulfation on tyrosine in nature is about 1%, mixtures of unsulfated $PSGL_{42-51}$ and sulfated $PSGL_{42-51}$–Y48s were prepared with a $PSGL_{42-51}$–Y48s/$PSGL_{42-51}$ ratio ranging from 1:1 to 1:100 (Supplementary Fig. 41). As shown in Supplementary Fig. 42, as the ratio decreased, the frequency of $PSGL_{42-51}$–Y48s decreased, whereas that of $PSGL_{42-51}$ remained constant. This observation aligns with the mixture preparation, in which only the concentration of $PSGL_{42-51}$–Y48s was changed. Based on the standard curve, the concentrations of $PSGL_{42-51}$–Y48s in the mixture were calculated to be 12.84 ± 0.89 μM, 1.23 ± 0.02 μM and 0.12 ± 0.03 μM, which are consistent with their prepared concentrations at the ratio of 1:1, 1:10 and 1:100, respectively. Notably, the $PSGL_{42-51}$–Y48s could not be detected by MS at the ratio of 1:100 (Supplementary Figs. 43 and 44). These results demonstrate the advantage of the nanopore in directly identifying and quantifying sulfated peptides in low abundance without purification or separation.

Distinguishing the isobaric sulfation and phosphorylation

Because the specific interactions allow the identification of sulfation, it is anticipated that these interactions could also be further used to distinguish sulfation from phosphorylation, a task that remains an ongoing challenge in MS due to their ultra-low difference in mass (~9.5 mDa). Tyrosine sulfation and phosphorylation were reported to both occur at Tyr165 of osteopontin (OST)[41]; thus, we chose the OST fragment (159-RGDSVVYG-166, $OST_{159-166}$) for our study. As shown in Fig. 5b, despite the slight differences in current blockade, the short duration of the OST peptides, particularly the phosphorylated peptide, resulted in a wide distribution of current blockade. This led to the low resolution of T232K mutant aerolysin for distinguishing sulfation and phosphorylation. The shortened duration induced by phosphorylation could be ascribed to its negative charges that suffer from strong electric field force under the applied voltage, thereby facilitating the translocation of the peptides. As the nanopore resolution is directly related to the duration of the molecule inside the pore[31], a net negatively charged region was constructed at the pore bottom to introduce an energy barrier for negatively charged sulfate group and phosphate group. As shown in Fig. 5a and Supplementary Fig. 45, with T232K mutation maintained, the positive charge of Lys238 near the pore bottom was mutated to neutral glutamine (K238Q), which exposed the negative charges of the originally paired Glu258. As expected, not only were the duration of $OST_{159-166}$–Y165s and $OST_{159-166}$–Y165p greatly prolonged by more than seven-fold (2.48 ± 0.14 ms and 2.06 ± 0.21 ms, respectively), but also the differences in current blockade between the fragments were enhanced to nearly 3 pA, leading to three well-separated populations in scatter plots (Fig. 5c–e and Supplementary Figs. 46 and 47). It should be noted that the shift in the current blockade of unmodified $OST_{159-166}$ between T232K and T232K/K238Q mutant aerolysin nanopores could be ascribed to the distinct open pore current of the nanopores (Supplementary Fig. 4).

Furthermore, the ability of T232K/K238Q mutant aerolysin was demonstrated to achieve the discrimination of sulfation without interference from phosphorylation in universal peptides. As shown in Fig. 5f–h and Supplementary Fig. 48, we evaluated the detection of the sulfation and phosphorylation on Tyr10 and Tyr14 of $CCR5_{7-16}$ (7-SPIYDINYYT-16) and on Tyr99 of $H3_{98-107}$ (98-AYLVGLFEDT-107) in the T232K/K238Q mutant aerolysin. For each site, the sulfation and phosphorylation induced well-separated current blockade with prolonged duration, distinguishable from the corresponding unmodified peptides. In addition, with T232K maintained, the mutation of K238Q was replaced by K238N to verify the nanopore design principle. As illustrated in Supplementary Fig. 49, three current peaks corresponding to $CCR5_{7-16}$, $CCR5_{7-16}$–Y10s and $CCR5_{7-16}$–Y10p, respectively, were clearly observed in the T232K/K238N mutant aerolysin consistent with that in T232K/K238Q, demonstrating the critical role of the exposed negative charges of Glu258, regardless of the neutral amino acids introduced at the Lys238 site. These observations highlight the importance of modulating interactions within the nanopore to regulate peptide transport processes, which, in turn, improves nanopore resolution. Therefore, by introducing the energy barrier for peptide transport, the specific interactions between the nanopore and tyrosine sulfation could indeed directly differentiate between isobaric sulfation and phosphorylation in universal peptides.

Discussion

In summary, we constructed an interaction-dependent nanopore to specifically map the tyrosine sulfation states in native peptides. Experiments with various nanopore mutations demonstrated the design principle for sulfation detection, and MD simulations confirmed the

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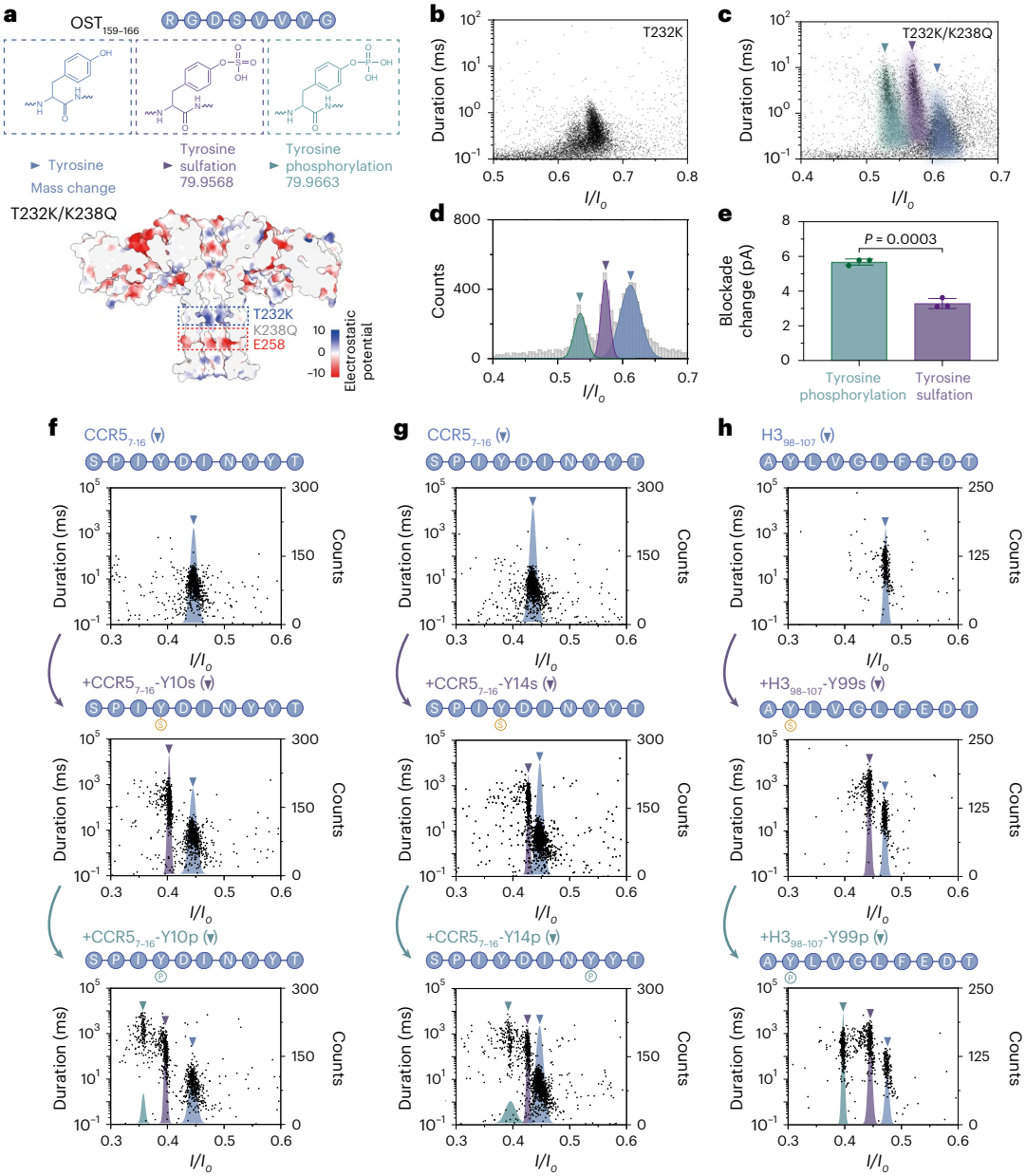


Fig. 5 | Discrimination of sulfation and phosphorylation on same tyrosine. **a**, Top, sequence of $OST_{159-166}$ and chemical structures of tyrosine, tyrosine sulfation and tyrosine phosphorylation. Bottom, charge density of the T232K/K238Q mutant aerolysin. The surface of the pore is colored according to its electrostatic potential as calculated by APBS (Adaptive Poisson-Boltzmann Solver) electrostatics calculations. **b**, Scatter plot of the peptide mixture at +120 mV by T232K mutant aerolysin. **c**,**d**, Scatter plot (**c**) and corresponding I/I_0 histogram with Gaussian fitting (**d**) of the peptide mixture at +120 mV based on T232K/K238Q mutant aerolysin, where the $OST_{159-166}$, $OST_{159-166}$–Y165s and $OST_{159-166}$–Y165p could be well discriminated. **e**, Blockade enhancement induced by sulfation and phosphorylation compared to the unmodified $OST_{159-166}$. Data are presented as mean ± s.d. for $n = 3$ replicates. Statistical analysis was conducted using unpaired two-tailed *t*-test, with *P* values shown. $P < 0.05$ was considered statistically significant. **f**–**h**, Discrimination of tyrosine, tyrosine sulfation and tyrosine phosphorylation in the Tyr10 site of $CCR5_{7-16}$ (**f**), the Tyr14 site of $CCR5_{7-16}$ (**g**) and the Tyr99 site of $H3_{98-107}$ (**h**).

critical role of non-covalent interactions on the determination. Our approach was demonstrated to be universal and unaffected by peptide sequences, which was hardly achieved in previous nanopore PTM studies[21,24,26]. A positive correlation between peptide mass and current blockade change induced by sulfation was further discovered. By using the instability of tyrosine sulfation, a framework was established, for the first time to our knowledge, in nanopore, to explore tyrosine sulfation in peptide mixtures and even in a native full-length protein. Moreover, our single-molecule method could directly measure the tyrosine sulfation at ultra-low abundance and without the interference from the isobaric phosphorylation in universal native peptides. We anticipate that this strategy could be instrumental in effectively and unambiguously discovering new tyrosine-sulfated proteins in wide dynamic ranges, further deepening the understanding of sulfation-related physiological and pathological processes.

Until now, MS has been the workhorse in the field of PTM identification. Nanopores, as an emerging single-molecule approach, could offer unique advantages, particularly in identifying low-abundance peptides and deciphering the chemical heterogeneity of PTMs, which are formidable challenges for MS. Nanopore studies have always been limited to detecting specific PTMs on certain peptides, but our sequence-independent strategy provides potential for practical

applications. However, the nanopore framework still faces a number of difficulties in identifying PTMs on unknown proteins in real samples. First, low protein concentrations, especially those with PTMs, pose a challenge in distinguishing the current signals of target molecules, which are often overwhelmed by background blockades. Apart from purifying the target analytes, precisely constructing the nanopore interface to enhance signal specificity and developing nanopore arrays to improve detection throughput could be keys to address this issue. Big data analysis with artificial intelligence technology, such as machine learning or deep learning, could further support PTM identification with high specificity. Second, the large volume and complex structures of proteins hinder their entering into nanopores. This challenge can be addressed by using enzymes to break down proteins into peptide fragments, akin to the bottom-up method in MS. By addressing these challenges, nanopores could become an ultra-sensitive, accurate and efficient method for PTM analysis on large scale. Additionally, integrating nanopore approach with MS is crucial. This synergy could enable the rapid and accurate identification of uncharacterized sulfated proteins and facilitate the discovery of new PTMs with low stability and short half-life—challenges often difficult to overcome with each method alone. Embracing a combination of multiple methods creates a win–win situation, which could promote advancements in PTM research and foster the prosperity of the proteomics field.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41589-024-01734-x.

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Methods

Chemicals and reagents

Peptides were purchased from GL Biochem and DG Peptide without further purification. The sulfated protein HIRV1 was supplied by C. He's group[43]. Tris(hydroxymethyl) aminomethane (≥99.9%, CAS no. 77-86-1) and EDTA (≥99%, CAS no. 60-00-4) were purchased from Aladdin. KCl (≥99.0%, CAS no. 7447-40-7), decane (≥99.8%, CAS no. 124-18-5) and trypsin-EDTA solution (no. T4049) were purchased from Sigma-Aldrich. 1,2-Diphytanoyl-*sn*-glycero-3-phosphocholine (DPhPC, no. 850356P) was purchased from Avanti Polar Lipids. Tryptone and yeast extract were purchased from Oxoid. DH5α *Escherichia coli* cells were purchased from TIANGEN Biotech. BL21(DE3) pLysS *E. coli* cells were purchased from TransGen Biotech. Isopropyl-β-D-thiogalactopyranoside (IPTG) was purchased from Thermo Fisher Scientific. Trifluoroacetic acid (TFA, ≥99.0%, CAS no. 76-05-1) was purchased Sigma-Aldrich. Methanol (≥99%, CAS no. 67-56-1) was purchased from Aladdin. Acetonitrile (≥99.9%, CAS no. 75-05-8) was purchased from Amethyst Chemicals. Endoproteinase Asp-N, sequencing grade, was purchased from Sigma-Aldrich. The primers used in this work were synthesized and HPLC-purified by Sangon Biotech. The aqueous solution in our experiments was prepared with ultrapure water (18.2 MΩ cm at 25 °C).

Pro-aerolysin gene mutation preparation

The primers used for plasmid mutation are provided in Supplementary Data 1. Pro-aerolysin gene mutants were prepared according to the overlap extension PCR method reported previously[44]. The PCR protocol was initiated by a 30-s denature step at 98 °C, followed by 30 cycles of denaturing (98 °C, 10 s), annealing (60 °C, 10 s) and extension (72 °C, 3 min, 30 s). Finally, the mixture was extended for 5 min at 72 °C. Products were then transformed into DH5α *E. coli* competent cells. Cells were grown on LB agar plates containing 100 μg ml^{-1} ampicillin overnight at 37 °C. Single clones were sent for sequencing (Supplementary Fig. 3).

Pro-aerolysin expression and purification

Plasmids were transformed into BL21(DE3) plysS chemically competent cells. Cells were grown on LB agar plates containing 100 μg ml^{-1} ampicillin overnight at 37 °C. A single colony was then harvested and inoculated into 15 ml of fresh LB medium containing 100 μg ml^{-1} ampicillin, and the culture was grown with 220-r.p.m. shaking at 37 °C overnight. The culture was diluted with 1 l of LB medium grown at 37 °C until the optical density at 600 nm (OD_{600}) reached approximately 0.8. Then, 0.5 mM IPTG was added to the media, and the culture was transferred to 20 °C for overnight growth with 220-r.p.m. shaking. The next day, the cells were centrifuged and then resuspended in 100 ml of lysis buffer (500 mM NaCl, 20 mM Na_3PO_4, pH 7.4) and crushed under a high-pressure cell crusher under 70 MPa. Afterwards, the crude lysate was centrifuged down at 4 °C for 30 min (~17,000*g*), and the supernatant was incubated with HisTrap HP His tag protein purification columns (Cytiva) at 2 min ml^{-1}. The target proteins were washed with elution and finally washed out by 20 mM Na_3PO_4, 0.5 M NaCl and 300 mM imidazole, pH 7.4. The molecular weight of the collected protein was confirmed by 12% SDS–PAGE.

Single-channel recording experiments and data analysis

Pro-aerolysin was incubated with trypsin-EDTA for 6 h at 25 °C. Lipid bilayer membranes were formed with DPhPC, which was dissolved in decane at a final concentration of 20 mg per 700 μl. The recording apparatus was separated by a 50-μm orifice (diameter), named *cis* and *trans* chambers. Then, 1 ml of 1.0 M KCl buffer was added into the two chambers, respectively. Two Ag/AgCl electrodes were immersed into the buffer and used to record the ionic current. Next, aerolysin protein was added to the *cis* chamber and could form a transmembrane heptameric nanopore. When a single aerolysin nanopore was inserted into the lipid, a constant current of nearly 50.0 pA could be seen under +100 mV at 24 ± 2 °C. Peptides were also added into the *cis* chamber. The current traces were amplified and recorded by using Clampex 10.7 software with a patch-clamp amplifier (Axon 200B) equipped with a Digidata 1440A A/D converter (Molecular Devices). The signals were acquired at a sampling rate of 100 kHz and filtered at 5 kHz. The data were analyzed using PyNanoLab 3.0.2 (ref. 45) and OriginLab 8.0 and 2023b (OriginLab Corporation) software.

Model constructions of MD simulations

Based on the cryogenic electron microscopy (cryo-EM) structure of wild-type aerolysin nanopore obtained from the Protein Data Bank (PDB ID: 5JZT)[46], a full-length model of wild-type aerolysin nanopore was constructed and inserted into a 200 Å × 200 Å path of dimyristoylphosphatidylcholine (DMPC) lipid bilayer. The lipid molecules overlapping with the protein nanopore were removed. Then, the nanopore–lipid system was solvated with pre-equilibrated TIP3P water molecules[47] using visual molecular dynamics (VMD)[48]. K^+ and Cl^- ions were added to neutralize the system and achieve a concentration of 1.0 M. The model was described by the CHARMM36 force field[49]. After a 10,000-step minimization and 2,000-step heating to 295 K, the system was equilibrated in the NPT ensemble at 295 K and 1 atm for 2 ns with the heavy atoms (non-hydrogen) of protein restrained, allowing the relaxation of lipid, water, ions and protein hydration atoms. Afterwards, the alpha carbons of the protein were still restrained, and the equilibration simulation lasted 10 ns. Next, all the restraints were removed, and the system was relaxed for 100 ns. All NPT ensemble simulations were carried out with a Langevin piston for pressure control and Langevin dynamics for temperature control[50]. Periodic boundary conditions were used in all directions. Van der Waals energies were calculated using a 12-Å cutoff, and the particle mesh Ewald (PME) method was used to treat long-range electrostatics[51]. After system equilibration, the T232K mutant aerolysin was constructed using the Mutator plugin in VMD. Additional 60-ns simulations with an electric field applied (+60 mV) were performed in an NVT ensemble without any positional restrain to model the experimental conditions.

The all-atom model of the $CCR2_{21–30}$ peptide was obtained from the AlphaFold Protein Structure Bank (AF-P41597-F1, residues 21–30)[33]. The peptides with PTMs were generated by patching a sulfate group to tyrosines of $CCR2_{21–30}$ peptide using the CHARMM36 force field[49]. Each peptide was solvated in a 60 Å × 60 Å × 60 Å water box with 1.0 M KCl. After a 10,000-step minimization and 2,000-step heating to 295 K, the peptide was equilibrated in the NVT ensemble at 295 K for 10 ns and in the NPT ensemble at 295 K for 20 ns with peptide backbone restrained. Next, all the restraints were removed, and the system was relaxed for 80 ns (Supplementary Fig. 13).

Constant-velocity SMD and MD simulations

After equilibration, the CCR2 peptides, including $CCR2_{21–30}$, $CCR2_{21–30}$-Y26s, $CCR2_{21–30}$-Y28s and $CCR2_{21–30}$-Y26/28s, were placed near the *cis*-entrance of the T232K mutant aerolysin with a certain position (−7.0, −9.0 and 80.0) for their mass center, respectively. The peptide was then pulled along the pore axis using the constant-velocity SMD[32]. Because the peptides were mainly driven by the EOF to enter the nanopore, the negatively charged amino acids and the sulfate group would be forced by the electric field. To better simulate the experimental situation, the backbone, the side chain of negatively charged amino acids and the sulfated group of the peptides were set as the SMD atoms in each case. These atoms were attached to a dummy atom via a virtual spring with a spring constant of 7.0 kcal mol^{-1} Å^{-2}. The dummy atom is moved to traverse the nanopore from *cis* to *trans* with a constant velocity of 0.005 Å ps^{-1}. Considering that the counter ions could potentially bind to the negative charges of peptides, leading to significantly low effective charges[36,37] (Supplementary Table 2), the SMD simulations were further conducted by attaching only the backbone of the peptides to the dummy atoms with the spring constant and constant velocity

remaining unchanged, to study the effects of the attachment on the simulation results.

The frames with the largest SMD force exerted on the peptides were used for further simulations to calculate the interactions between the peptides and nanopore. The model was simulated at an electric field of + 60 mV in an NVT ensemble for 80 ns with harmonic constraints imposed on the backbone of the peptide. Based on the trajectories from MD simulations, the NAMD Energy plugin 1.4 in VMD was employed to calculate the van der Waals and electrostatic interactions between the peptides and nanopore. Two selections of the peptide and aerolysin were set with a 12-Å cutoff to calculate the interactions between those selections based on each snapshot (2 ps) of 2-ns MD trajectory within the final 20 ns of the 80-ns simulation. During the same time period, hydrogen bonds between the peptides and nanopore were calculated with the donor–acceptor distance (D···A distance) shorter than 3.0 Å and the H–D···A angle lower than 20° (ref. 28). To validate the results, another relaxed parameter, with the D···A distance shorter than 3.5 Å and the H–D···A angle less than 30°, was also tested for calculating hydrogen bonds between the peptides and nanopore. The CHARMM36m force field[52] was further used to describe the aerolysin system, with all other parameters kept the same, to validate the enhanced non-covalent interactions between the peptide and nanopore induced by tyrosine sulfation (Supplementary Figs. 16 and 17).

Calculation of effective charge

$$\tau = \tau_0 \cdot \exp\left(-\frac{V}{V_c}\right)$$

$$V_c = \frac{k_B T}{Z_{\text{inside}} e}$$

The effective charge (Z_{inside}) of the peptide inside nanopore is estimated using the above equation with the exponential fitting of the voltage-dependent duration. In the formula, τ represents translocation duration; τ_0 is a diffusive relaxation time; V is applied voltage; V_c is the critical potential needed to overcome the free energy barrier for squeezing a peptide to translocate through the confined nanopore; e is the magnitude of the elementary charge; k_B is the Boltzmann constant; and T is the thermodynamic temperature.

Calculation of enhancement in current blockade

$$\text{Blockade change } (B_{\text{sulfated}}) = \text{Current}_{\text{baseline}}(\text{I/I}_{0\ \text{sulfated}} - \text{I/I}_{0\ \text{unmodified}})$$

$$\text{Average blockade change} = (B1_{\text{sulfated}} + B2_{\text{sulfated}})/2$$

The changes in I/I_0 between sulfated and unmodified peptides were calculated using the mixture data obtained from the I/I_0 histograms, which were fitted with Gaussian distribution. $\text{Current}_{\text{baseline}}$ represents the baseline current. The average blockade change is calculated as the average blockade change (B_{sulfated}) for the two single-site sulfated peptides, except for $H3_{98-107}$.

Procedures of tyrosine desulfation

Sulfated tyrosine is susceptible to losing its sulfate group when exposed to an acidic environment and heat[53]. To induce this desulfation, the peptide solution was acidified by using TFA to a final concentration of 0.5% (v/v). Then, the peptide–TFA solution was incubated at 50 °C for 8 h. Finally, the mixture was neutralized with sodium hydroxide solution before the measurement with the nanopore.

HIRV1 protein sequence and digestion

The amino acid sequence of full-length HIRV1 is VVYTDCTESGQNLCLCEGSNVCGQGNK CILGSDGEKNQCVTGEGTPKPQSHNDGDFEEIPEEYLQ, where the tyrosine sulfation should occur at Tyr63. During the protein synthesis, the biotin modification and Bpa were performed at Lys47 and Phe56, respectively, giving its final structure of VVYTDCTESGQNLCLCEGSNVCGQGNKCILGSDGEKNQCVTGEGTP**K(Biotin)**PQSHNDGD**Bpa**EEIPEE**sY**LQ[43].

A total of 45 µg of HIRV1 protein was taken, and the volume was adjusted to 50 µl using 10 mM Tris-HCl buffer (pH 7.5). Endoproteinase Asp-N was reconstituted by dissolving the lyophilized enzyme in 100 µl of 10 mM Tris-HCl buffer, pH 7.5, to give a final concentration of 0.02 µg μl^{-1}. Then 10 µl of endoproteinase Asp-N was added at a 1:225 (w/w) enzyme-to-protein ratio, and the sample was subsequently incubated overnight at 37 °C. The samples were stored at −20 °C before use.

MS analysis

The peptide solution was loaded onto a C18 ZipTip (Millipore) that had been washed with 100% acetonitrile and equilibrated with 0.1% TFA. The ZipTip was then washed with 0.1% TFA, and the sample was eluted in 75% acetonitrile containing 0.1% TFA. The desalted samples were identified and quantified using an Orbitrap Q Exactive high-resolution mass spectrometer (Thermo Fisher Scientific) equipped with a heated electrospray ionization (HESI) source. The following generic MS tune parameters were used: spray voltage of 3.5 kV; capillary temperature of 320 °C; sheath gas (nitrogen) flow of 35 a.u.; auxiliary gas (nitrogen) flow of 10 a.u.; and probe heater temperature of 400 °C. A full MS scan was acquired for each run in the range of m/z 200–2,000 with a resolving power of 140,000 full width at half maximum. The automatic gain control target value and maximum injection time used for the full MS scan were 3×10^6 and 50 ms, respectively. The raw data obtained from ESI-Quadrupole-Orbitrap MS were processed using Xcalibur software version 4.1 (Thermo Fisher Scientific).

Capture rate calculation

The capture rate (frequency) was calculated with the total event number (n_{analyte}) and its effective time (τ_{efftive}). Here, τ_{efftive} was calculated with the following equation:

$$\tau_{\text{efftive}} = \tau_{\text{total}} - \sum \tau_{\text{off analyte}}$$

τ_{total} represents the total recording time. Then, the frequency (f) of each peptide was calculated using the following formula:

$$f = n_{\text{analyte}}/\tau_{\text{efftive}}$$

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data supporting the findings of this study are available in the main text and the Supplementary Information. Additional raw data and the aerolysin cryo-EM structure are available at figshare (https://doi.org/10.6084/m9.figshare.26494222). Source data are provided with this paper.

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Author contributions

M.-Y.L., H.N. and Y.-T.L. conceived the idea and designed the experiments. H.N. performed the nanopore experiments. H.N. and Y.G. analyzed the nanopore data. H.N. constructed the MD simulations, with assistance from M.-Y.L. H.N., J.-G.L., J.J. and Y.-L.Y. expressed and purified protein nanopores. H.N. and M.-Y.L. interpreted the data and wrote the manuscript. M.-Y.L. and Y.-T.L. supervised the project. All authors critically reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Software and code

Policy information about availability of computer code

Data collection	Single-channel recording data were acquired by using Clampex 10.7 software (Axon Instruments). Molecular Dynamics simulations were carried out by using NAMD 2.13 program.
Data analysis	Current traces and current blockages were analyzed by using pynanolab 3.0.2. OriginLab 8.0 and 2023b software were used for scatter plots, histograms construction, curve fitting and graph presentation. MD simulations results were visualized and analyzed with VMD 1.9.4. The raw data obtained from ESI-Quadrupole-Orbitrap MS were processed using Thermo Xcalibur software version 4.1. Adobe Illustrator 2023 was used for making figures.

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Data supporting the findings of this study are shown in the main text and the supplementary information. All the data supporting the findings of this study are available at https://doi.org/10.6084/m9.figshare.26494222.
The peptide and protein structures can be obtained from Protein Data Bank (5JZT, https://www.rcsb.org/structure/5JZT) and AlphaFold Protein Structure Bank (AF-P41597-F1, https://alphafold.ebi.ac.uk/entry/P41597).

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Sample size	Nanopore measurements were conducted continuously to capture all events during the measurement. Each sample consists of all events recorded under a specific condition. For scatter plots, each sample included a minimum of 300 events, providing a well-defined population sufficient enough for statistical analysis.
Data exclusions	No data were excluded from this study.
Replication	All replication attempts were successful. The reproducibility of the experimental findings was verified in at least three independent experiments unless otherwise stated.
Randomization	Randomization is not applicable to this study, as peptide samples do not need to be assigned to experimental groups.
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